

Date: _____

2) CA - Assignment of
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Roll-no: 24k-0737

Q#1:-

Data:- P_2 takes 25% less time but incurs 20% more CPI.
clock-frequency of P_1 is 1 GHz
clock-frequency of P_2 is 1.6 GHz

Sol:-

$$T_2 = 0.75 T_1$$

$$CPI_2 = 1.2 CPI_1$$

$$\frac{T_2}{T_1} = \frac{CPI_2 / f_2}{CPI_1 / f_1} \quad f_1 = 1 \text{ GHz}$$

$$0.75 = \frac{1.2 CPI_1 / f_2}{CPI_1 / 1} \quad 2) \quad 0.75 = 1.2 / f_2$$
$$f_2 = \frac{1.2}{0.75} = 1.6$$

$$[f_2 = 1.6 \text{ GHz}] \text{ Ans.}$$

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Q42:- Data:-

MTTF = 2000000 hrs

Servers = 25000

SSDs / server = 4

Total SSDs = 25000 x 4 = 100000 SSDs.

(a) fit for a single SSD:-

$$\frac{FIT}{MTTF} = \frac{10^9}{2000000} = 500 \Rightarrow [FIT = 500] \quad \text{Ans}$$

(b) failure-rate for the entire WSC:-

$$\lambda = \frac{1}{MTTF} = \frac{1}{2000000}$$

$$MTTF = 2000000$$

$$\lambda_{WSC} = 100000 \times \frac{1}{2000000}$$

$$\lambda_{WSC} = 0.05 \text{ failures/hr} \quad \text{or} \quad 0.05 \times 10^9 = [50,000,000 FIT] \quad \text{Ans}$$

(c) MTTF for entire WSC:-

$$MTTF_{WSC} = \frac{1}{\lambda_{WSC}} = \frac{1}{0.05} \Rightarrow [20 \text{ hrs}] \quad \text{Ans}$$

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(d) New individual (SD) MTTF :- 1500000 hrs

total SDs = 100,000

System-rate / failure-rate = $\lambda_{ws} = \frac{100000}{15000000} = 0.06667$

MTTF_{ws} = $\frac{1}{0.06667}$ s/15 hours / *Am*

(e) Since the MTTF decreases from 2000000 hrs to 1500000 hrs, reliability doesn't improve. The new MTTF is 0.75x the original, while the failure rate becomes approximately 1.33x higher. *Am*

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Q#3:

Given:-

Instruction-type	frequency	CPI
integer-Arithmetic	40%	1
Load	25%	3
store	25%	3
FPSQRT	10%	10

(a) Overall CPI = ?

Sol:-

$$CPI = (0.40 \times 1) + (0.25 \times 3) + (0.25 \times 3) + (0.10 \times 10)$$

$$= 0.40 + 0.75 + 0.75 + 1.00$$

$$[CPI = 2.90 \text{ or } 2.9] \quad \text{Ans.}$$

(b) Clock-speed = 3GHz

Instruction-Count = 20 billion

CPU-time = ?

Sol:-

$$CPU\text{-time} = \frac{IC \times CPI}{\text{Clock-rate}}$$

clock-rate

$$= \frac{30 \times 10^9 \times 2.9}{3 \times 10^9} = 87/3 = 29$$

$$[CPU\text{-time} = 29 \text{ sec}]$$

Ans.

Q44:-

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(a) Ans The faster growth was caused by improvements in both technology & processor-architecture.

- Transistors became smaller & more numerous, allowing processors to operate at higher frequencies & implement more advanced architectures including:
 - (i) pipelining.
 - (ii) Superscalar execution.
 - (iii) instruction level execution.

(b) 52% → 23% → 12% → 3.5% (Performance after 2003).

Several factors caused this slowdown, including:-

- (i) Clock-frequency limitations
 - (ii) Power-consumption.
 - (iii) Memory-Wall
 - (iv) limited instruction-level parallelism.
 - (v) Leakage-power.
 - (vi) Shift toward multicore-processors
- Ans.

(c)

Dynamic scaling allowed transistor size to decrease while voltage & power consumption also decreased & the break-down caused excessive power-consumption & heat & this created Power wall which limited further frequency increase & slowdown the performance-growth. *Ans*

(d) After power-wall, increasing frequency causes:

- (i) High power consumption
- (ii) More heat
- (iii) Greater cooling effort requirement
- (iv) Increased leakage power
- (v) Reliability-problems. *Ans*

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Q#5.

(a)

Standardized benchmarks are needed because different processors have different:-

- (i) Clock-frequency
- (ii) Number of cores
- (iii) Cache-size
- (iv) Architecture
- (v) Memory-systems

→ Standardized benchmarks provide a common workload & methodology. *Ans.*

(b) SPECint: measure performance on integer-based workloads.
SPECfp: measure performance on floating point workloads. *Ans.*

(c) SPEC benchmarks provide a fair comparison as it uses:

- (i) Benchmark-program
 - (ii) Input-data
 - (iii) Measurement-procedure
 - (iv) Compilation & Execution-rules
 - (v) Reporting-method
- Ans.*

(d)

A suite of benchmark is generally better because computers run many different types of applications. A single program may favor a particular architecture, whereas multiple benchmarks provide a broader & more representative evaluation of processor performance.

(e)

A higher SPEC performance score indicates better performance on the benchmark workload meaning the processor or system can execute those workloads faster relative to the reference system.

(f) One limitation of benchmark is that they can't perfectly represent every real world work-load. A ~~prof~~ processor that perform very well on a benchmark may not perform equally well for every application.